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Linear guide device

Abstract

A linear guide device having a guide carriage with two guide units, each guide unit including two guide modules abutting guide surfaces of a base structure in an operating position of the guide carriage. One of the two guide modules is an adjustable guide module, which is adjustable by means of one of two adjustment units to set a pretension between the guide modules and the guide surfaces. For synchronous actuation of the two adjustment units, a central actuating device is provided which is drivingly coupled to both adjustment units.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to German application 10 2022 120 154.6, filed Aug. 10, 2022, which is incorporated by reference.

BACKGROUND

(2) The invention relates to a linear guide device with a guide carriage featuring a carriage base body, which has two guide units arranged one after the other in a carriage longitudinal direction, each with a first guide module arranged on the carriage base body and a second guide module arranged opposite the first guide module on the carriage base body at a distance in a carriage transverse direction orthogonal to the carriage longitudinal direction, wherein the guide carriage, in its operating position, is supported linearly movably on a base structure in the carriage longitudinal direction, wherein the two first guide modules bear against a first guide surface and the two second guide modules bear against a second guide surface of the base structure so as to be movable with

respect thereto, wherein one of the two guide modules of each guide unit is an adjustable guide module and each of the two guide units is provided with one of two adjustment units arranged successively in the carriage longitudinal direction, by means of which the associated adjustable guide module is admittable by an adjusting force to cause an adjustment which is oriented in the carriage transverse direction, relative to the carriage base body, wherein a pretension is adjustable with which the two guide modules of the guide unit are pressed against the respectively associated first or second guide surface of the base structure in the operating position of the guide carriage.

(3) A linear guide device of this type known from DE 10 2020 204 569 A1 has a guide carriage which is supported linearly movably on two mutually facing first and second guide surfaces of a base structure having a longitudinal extension. The movement direction coincides with a carriage longitudinal direction of the guide carriage. For the support, the guide carriage can have two guide units arranged on a carriage base body and arranged in succession in the carriage longitudinal direction. Each guide unit has a first guide module arranged stationary on the carriage base body and a second guide module lying opposite the first guide module in a carriage transverse direction, the second guide module being an adjustable guide module whose relative position assumed with respect to the carriage base body in the carriage transverse direction can be variably adjusted by carrying out an adjusting movement. In the operating position of the guide carriage, the two first guide modules are pressed against the first guide surface and the two adjustable second guide modules are pressed against the second guide surface of the base structure. Each guide unit has its own adjustment unit which makes it possible to apply an adjustment force to the adjustable guide module in order to cause the adjusting movement and to set a pretension with which the two guide modules of the respective guide unit are pressed against their respective guide surface of the base structure. Each adjustment unit contains a displacement medium arranged in the carriage base body, which can be subjected to a displacement force by an actuator in order to generate the adjustment force acting on the adjustable guide module. The adjusted pretension has a decisive effect on the precision with which the guide carriage is guided on the base structure during its intended use.

(4) The presence of two axially spaced separate guide units offers the advantage of being able to absorb relatively high lateral forces as well as tilting forces and torques acting on the guide carriage. In this context, the set pretensions of the two guide units should lie in a comparable range. However, the adjustment of the two guide units to each other in this respect is relatively complex.

(5) Pretension elements are known from EP 0 142 069 A2, with which individual guide modules designed as circulation shoes can be individually adjusted with respect to their pretension in relation to a longitudinal guide. Each guide module is assigned its own pretensioning element. The pretension element has a housing which is supported on the guide module and in which two parts are arranged which can be adjusted by means of a tensioning screw and which interact with wedge surfaces of a support plate arranged on the guide carriage.

SUMMARY OF THE INVENTION

(6) The invention is based on the problem of taking measures which enable precise adjustment of the pretension acting between the two guide units and the associated guide surfaces in a simple manner.

(7) To solve this problem, it is provided in accordance with the invention, in conjunction with the features mentioned at the beginning, that the guide carriage features a central actuating device drivingly coupled to the adjustment units of both guide unit, through the actuation of which synchronous adjusting forces can be exerted on the two adjustable guide modules.

(8) In this way, there is the advantageous possibility of adjusting the pretension, also referred to as guide pretension, between the guide modules of the two guide units and the associated guide surfaces of a base structure by means of a uniform actuation process, and in this way to obtain a pretension of comparable intensity for all guide modules with little effort. In this way, the forces acting on the guide carriage during operation can be absorbed jointly by the guide modules of both

guide units, resulting in a high load capacity of the linear guide measures. By means of the central actuating device, the adjustment units of the two guide units are drive-coupled to each other, so that actuation of the central actuating device results in simultaneous actuation of both adjustment units, with the result that synchronous adjusting forces are generated with respect to the two adjustable guide modules. The drive coupling allows mutual feedback between the two adjustment units, so that the pretension forces can be distributed evenly and ultimately, with a single adjustment process, both guide units can be easily adjusted so that the guide carriage is precisely guided in its operating position and can absorb high lateral forces and torques with low susceptibility to wear. The assembly time for the linear guide device is significantly reduced by the measures according to the invention compared to the prior art, so that a cost-effective production of the linear guide device is possible.

(9) Advantageous further embodiments of the invention can be seen from the dependent claims.

(10) Preferably, all guide modules are designed as rolling-element guide modules, which rest with rolling bearing elements on the respectively assigned guide surface of the base structure. When the guide carriage moves relative to the base structure, the rolling bearing elements may roll on the associated guide surface. The rolling-element guide modules are preferably designed as recirculating ball guide modules, which advantageously each feature at least one internal recirculating channel in which a plurality of spherical rolling bearing elements are accommodated, which may recirculate in the recirculating channel in the channel longitudinal direction. It is advantageous that each guide module contains two circulation channels arranged adjacent to one another with rolling bearing elements accommodated therein.

(11) Each adjustment unit is advantageously arranged in the carriage transverse direction between the first guide module and the second guide module of the associated guide unit. This allows a compact construction. Furthermore, in this manner, the two adjustment units can be coupled to each other via the central actuating device by a short distance/route, i.e., a short connection distance.

(12) In a particularly preferred embodiment, the two adjustment units are each designed as a wedge gear. An actuating force effective in the carriage longitudinal direction can be introduced into each wedge gear via the central actuating device, which can be converted by the wedge gear into an adjusting force effective orthogonally to the carriage longitudinal direction, which acts on the adjustable guide module of the respective guide unit. The design of the respective wedge gear can be used to achieve a desired force transmission ratio that makes it possible to generate high setting forces from a relatively low actuating force. Wedge gears as such are known in various designs, for example from DE 10 2011 006 323 A1.

(13) The two wedge gears are advantageously designed so that their actuating directions are opposite to each other. The actuating forces generated by the central actuating device are introduced into the two wedge gears in opposite directions in the carriage longitudinal direction. Preferably, each wedge gear has a wedge-shaped adjusting element that can also be referred to as wedge element, with the two wedge-shaped adjusting elements being simultaneously pushed away from each other, i.e., apart, when the central actuating device is actuated. However, the wedge gears may also be designed in such a way that the two wedge-shaped adjusting elements can be acted upon simultaneously by the central actuating device in a direction towards each other in order to generate the respective adjusting force.

(14) In an advantageous further development, each of the two adjustment units has a movable adjusting element drivable relative to the carriage base body for a driving movement in the carriage longitudinal direction and an with respect to the carriage base body in the carriage longitudinal direction immovable adjusting element. The immovable adjusting element immovable in the carriage longitudinal direction, which for simplification purposes will also be referred to below simply as the “immovable adjusting element”, is arranged in an absolutely immovable manner on the carriage base body in an advantageous embodiment and in particular is formed integrally with the carriage base body.

(15) In both adjustment units, the movable adjusting element is force-transmittingly coupled to the adjustable guide element for exerting the adjusting force in the carriage transverse direction.
(16) Furthermore, in each adjustment unit, each of the two adjusting elements belonging thereto is provided with one of two force-deflecting surfaces abutting against each other in a relatively slidable manner, such that they can slide relative to one another, of which at least one is in the form of an oblique surface inclined with respect to the carriage longitudinal direction and to the carriage transverse direction. When a movable adjusting element executes a driving movement in the carriage longitudinal direction as a result of an actuation of the central actuating device, the force deflection surface formed on it slides on the force deflection surface of the immovable adjusting element, resulting in the adjusting movement of the associated adjustable guide module oriented transversely to the driving movement.
(17) The driving movement of the movable adjusting element and the adjusting movement of the adjustable guide module take place in each guide unit as simultaneously overlapping movements.
(18) Preferably, the movable adjusting element of each adjustment unit features a pressing surface facing the adjustable guide module, in particular in the carriage transverse direction. With this pressing surface, the movable adjusting element rests against a force-receiving surface of the adjustable guide module, wherein the adjustable guide module is supported immovably with respect to the carriage base body in the carriage longitudinal direction, so that it is movable exclusively in the carriage transverse direction. During the driving movement of the movable adjusting element, the movable adjusting element slides with its pressing surface in the carriage longitudinal direction against the force receiving surface of the adjustable guide module, so that the axial relative position between the movable adjusting element and the adjustable guide module changes, wherein the movable adjusting element simultaneously exerts the adjusting force on the adjustable guide module due to a superimposed transverse movement. In particular, the pressure surface is designed and arranged in such a way that a normal vector perpendicular to the pressing surface is aligned in the same way in the carriage transverse direction as a normal vector of the force-absorbing surface.
(19) Of the two mutually abutting force deflection surfaces, only one may in principle be formed as an oblique surface, while the other may be formed on one or more cam-like projections so that it only abuts the oblique surface at certain points. In terms of optimum force transmission and low-wear design, however, it is advantageous if each of the two force deflection surfaces abutting each another is designed as an oblique surface inclined to the longitudinal axis of the carriage, i.e., within a respective adjustment unit with the same inclination. In this way, large area of contact between the force deflection surfaces abutting each other can be ensured.
(20) It is considered advantageous if the inclination of the at least one force deflection surface of one adjustment unit is opposite to that of the at least one force deflection surface of the other adjustment unit. In this case, the two movable adjusting elements can be driven by the central actuating device to mutually oppositely oriented driving movements with respect to the carriage longitudinal direction in order to generate the adjusting forces acting on the adjustable guide module. In particular, the inclinations are aligned in such a way that the adjustment forces result from driving movements of the movable adjustment units directed away from each other, i.e., with movable adjustment elements moving away from each other in the carriage longitudinal direction.
(21) In principle, the central actuating device may be designed to be actuated by a motor and may feature, for example, an electric drive unit or a drive unit actuated by fluid power and in particular pneumatically, which in particular also permits remote actuation. However, since the desired pretension usually must be adjusted only once, namely during assembly of the linear guide device, it is considered more favorable if the central actuating device is of a manually actuatable design and has at least one actuating member for manual actuation.
(22) At least one actuating member can be assigned to one of the two axial end faces of the guide carriage so that a good accessibility is provided. Preferably, the central actuating device is equipped

with two actuating members, which are assigned in particular to opposite axial end regions of the guide carriage and which can be used alternatively to set the desired pretension. The at least one actuating member is particularly a screw member.

(23) A particularly simple and compact adjustment mechanism is characterized by the fact that the central actuating device for drive coupling with the two adjustment units features a rod-shaped coupling structure, referred to as a rod coupling structure, which, with compact dimensions, allows opposing actuating forces to be introduced synchronously into the two movable adjusting elements of the two adjustment units in the carriage longitudinal direction in order to generate the adjustment forces of the two adjustment units. The rod coupling structure can bridge a distance existing between the two adjustment units in order to engage the two movable adjusting elements.

(24) It is convenient if the rod coupling structure features a coupling rod extending between the two movable adjusting elements, bridging the distance between the two movable adjusting elements, while being axially supported with respect to each movable adjustment element. On at least one of the two movable adjusting elements, a rotatable actuating member of the rod coupling structure cooperating with the coupling rod is arranged, on which the coupling rod is axially supported with respect to the respective movable adjusting element. A rotary actuation of the at least one actuating member may cause a change in distance between the two movable adjusting elements, with which the driving movements of the two movable adjusting elements are accompanied.

(25) For example, the coupling rod may be fixed to one of the two movable adjusting elements and only cooperate with a rotatable actuating member fixed to the other movable adjusting elements, which is formed separately in this respect.

(26) A particularly advantageous embodiment is considered to be one in which the rod coupling structure features in each movable adjusting element a threaded hole extending in the carriage longitudinal direction, into which the coupling rod extends axially displaceable with one of its two end sections, wherein an actuating member formed as a screw member is screwed into each threaded hole, on which the coupling rod with its associated end section is supported at the end face and which actuating member is axially adjustable by rotation relative to the associated movable adjusting element. Preferably, each threaded hole is provided with an internal thread only over a partial length in which the screw member extends. A threadless length section of the threaded hole may be provided to receive the associated end section of the preferably cylindrically formed coupling rod in a linearly displaceable guided manner.

(27) Preferably, once the pretension has been set, i.e., in the operating position of the guide carriage, the adjustable guide modules are stationary fixed to the carriage base body. For this purpose, a fastening device can be assigned to each adjustable guide module, for example in the form of a screw fastening device with one or two clamping screws. The pretension is set when the fastening devices are deactivated and fixed unchangeably after setting by activating the fastening devices. In particular, the fixation can be detachable so that readjustments of the pretension are possible if required. In a design as a screw connection device, the deactivated state corresponds to a loosened screw connection and the activated state to a tightened screw connection.

(28) The linear guide device advantageously contains the base structure already mentioned, wherein this base structure features a longitudinal extension and has the first and second guide surfaces already mentioned. The two guide surfaces are arranged at a distance from one another. Preferably, the two guide surfaces face each other, wherein the two guide units are arranged at least partially in a clearance of the base structure present between the two guide surfaces. The guide carriage is supported linearly movably on the base structure in the longitudinal direction thereof, in that it bears movably against the first guide surface with its two first guide modules and against the second guide surface with its two second guide modules. Guide modules designed as rolling-element guide modules rest with their rolling bearing elements in a rollable manner against the respectively assigned guide surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention is explained in more detail below with reference to the accompanying drawing. Showing:

(2) FIG. **1** a preferred embodiment of the linear guide device according to the invention in an isometric representation with a guide carriage supported linearly movably on a base structure,

(3) FIG. **2** a side view of the linear guide device with viewing direction according to arrow II in FIG. **1**,

(4) FIG. **3** a half cross-section of the linear guide device according to cut line III-III in FIG. **2**, wherein only the left half of the image is partially cut, and the right half of the image is uncut,

(5) FIG. **4** an axial front view of the linear guide device with the direction of view according to arrow IV in FIG. **2**,

(6) FIG. **5** a stepped longitudinal cut of the linear guide device with a view at the guide carriage according to cutting line V-V in FIG. **3**,

(7) FIG. **6** a further stepped longitudinal cut of the linear guide device looking with a view from below at the base structure according to cut line VI-VI in FIG. **3**,

(8) FIG. **7** an isometric exploded view of the linear guide device,

(9) FIG. **8** an isometric bottom view of the guide carriage without representation of the base structure combined with the guide carriage in the operating position, and

(10) FIG. **9** a perspective view of the guide carriage of the linear guide device analogous to FIG. **8**, but with a longitudinal cut in the area of the guide units according to cut plane IX-IX in FIG. **4**.

DETAILED DESCRIPTION OF THE INVENTION

(11) The linear guide device, designated overall by reference sign **1**, has a longitudinal extension along an imaginary longitudinal axis **2** and also has a transverse axis **3** perpendicular to the longitudinal axis **2** and a vertical axis **12** orthogonal to the longitudinal axis **2** and the transverse axis **3**.

(12) One of several components of the linear guide device **1** is a base structure **4** having a longitudinal extension and extending along the longitudinal axis **2**. The base structure **4** has a base body **5** exemplarily having a U-shaped cross-section, on which a first guide surface **6** and a second guide surface **7** opposite the first guide surface **6** at a distance in the axial direction of the transverse axis **3** are formed. The two guide surfaces **6**, **7** each have a strip-shaped longitudinal form and extend parallel to the longitudinal axis **2**.

(13) The base body **5** has two leg sections **8***a*, **8***b* opposite each other in the axial direction of the transverse axis **3**, which longitudinally delimit a base body intermediate space **9** on opposite sides. On the inner sides of each leg section **8***a*, **8***b* facing the base body clearance **9**, one of the two first and second guide surfaces **6**, **7** is formed, which thus face each other.

(14) The linear guide device **1** has an upper side **13** facing in the axial direction of the vertical axis **12** and a lower side **14** opposite thereto. The gutter-shaped base body clearance **9** is open longitudinally towards the upper side **13**.

(15) The linear guide device **1** includes a guide carriage **15** which, in its operating position as shown in FIGS. **1** to **6**, is supported linearly movably on the base structure **4** in the axial direction of the longitudinal axis **2**. Unless otherwise specified in an individual case, the following description refers to this operating position.

(16) The guide carriage **15** has a carriage longitudinal axis **16** which extends parallel to the longitudinal axis **2** and whose axial direction is designated as carriage longitudinal direction **16***a* for better differentiation. The linear movement **17**, indicated by a double arrow, executable by the guide carriage **15** with respect to the base structure **4**, is executable in the carriage longitudinal direction **16***a* as arbitrarily reciprocating movement.

(17) The guide carriage **15** further has a carriage transverse axis **18** extending in a carriage transverse direction **18***a* orthogonal to the carriage longitudinal direction **16***a*, and a carriage vertical axis **19** extending in a carriage vertical direction **19***a* orthogonal to the carriage longitudinal direction **16***a* and to the carriage transverse direction **18***a*. The longitudinal axis **2** and the carriage longitudinal axis **16**, furthermore the transverse axis **3** and the carriage transverse axis **18**, and finally the vertical axis **12** and the carriage vertical axis **19** are respectively parallel to each other.
(18) The guide carriage **15** is equipped with two guide units **22**, **23** arranged in succession in the longitudinal direction **16***a* of the carriage, which are in the following also referred to as the first guide unit **22** and the second guide unit **23** for better differentiation. The two guide units **22**, **23** are in particular components of a lower end section **24** of the guide carriage **15**, which extends into the base body clearance **9** from the upper side **13**. The two guide units **22**, **23** are conveniently spaced apart from each other in the carriage longitudinal direction **16***a*.
(19) Each of the two guide units **22**, **23** includes a first guide module **25** and a second guide module **26** spaced with respect thereto in the carriage transverse direction **18***a*. With respect to the carriage transverse direction **18***a*, the two first guide modules **25** are located on one side of the central carriage longitudinal axis **16**, while the two second guide modules **26** are located on the opposite side of the carriage longitudinal axis **16**.
(20) The two first guide modules **25** and the two second guide modules **26** are each arranged in succession at a distance from each other in the longitudinal direction **16***a* of the carriage.
(21) The two first guide modules **25** rest movably against the first guide surface **6** in this respect, while the two second guide modules **26** bears movably against the second guide surface **7** in this respect. During the linear movement **17**, the guide modules **25**, **26** move along the respective associated guide surface **6**, **7**, wherein the contact between the guide modules **25**, **26** and the guide surfaces **6**, **7** results in a transverse support of the guide carriage **15** with respect to the base structure **4** in the carriage transverse direction **18***a*, so that an exact linear movement **17** of the guide carriage **15** is possible.
(22) During operation of the linear guide device **1**, an object to be guided is usually fixed to the guide carriage **15**, for example a component of a machine. For fastening the object, the guide carriage **15** is equipped with at least one fastening interface **29**, which are exemplarily designed as fastening holes. During operation of the linear guide device **1**, transverse forces introduced into the guide carriage **15** by the attached object are absorbed by the base body **5** through the guide contact between the guide modules **25**, **26** and the guide surfaces **6**, **7**.
(23) Exemplarily, the base structure **4** also absorbs torques and tilting moments, which can be attributed to a design and profiling of the guide modules **25**, **26** and the guide surfaces **6**, **7** that can be seen in FIGS. **3** and **4**.
(24) Each guide module **25**, **26** has at least one guide element **27** with which it rests against the associated guide surface **6**, **7** with support. According to an unillustrated embodiment, the guide element **27** is a sliding element which slides along the associated guide surface **6**, **7** during the linear movement **17**. Preferably and in accordance with the illustrated embodiment, all guide modules **25**, **26** are designed as rolling-element guide modules, each of which has a plurality of guide elements **27**, which are designed as rolling bearing elements **27***a*, which rollable rest on the respectively associated guide surface **6**, **7**. The rolling bearing elements **27***a* can, for example, be of roller-shaped or spherical design.
(25) Exemplarily, the rolling bearing elements **27***a* are spherical in shape, wherein a preferred characteristic is that the guide modules **25**, **26** are designed as recirculating ball guide modules.
(26) Each guide module **25**, **26** has a housing, in particular a multi-part housing, designated as a guide housing **28**, in which two closed circulation channels **31** are formed, lying one above the other in the carriage vertical direction **19***a*, in which in each case a multiplicity of rolling bearing elements **27***a* are accommodated in a row, which during the linear movement **17** execute a circulation movement **32**, indicated by a double arrow, in the associated circulation channel **31**.

Each circulation channel **31** has, on the longitudinal side facing the guide surface **6**, **7**, a slot-like opening through which the rolling bearing elements **27***a*, which are currently located in its area, may partially project in order to be in guide contact with the opposite first or second guide surface **6**, **7** and to roll thereon while executing the circulation movement **32**.
(27) The rolling bearing elements **27***a* lie outside the sectional plane in FIGS. **5** and **6** and are therefore indicated by dashed lines only in FIG. **6** in one of the two second guide modules **26**.
(28) Instead of two recirculation channels **31**, the recirculating ball guide modules can also have only a single recirculation channel **31** with rolling bearing elements **27***a* according to an unillustrated embodiment.
(29) The guide carriage **15** has a carriage base body **33**, which is preferably formed in one piece and in particular is made of metal. The guide modules **25**, **26** are designed separately with respect to the carriage base body **33** and are each fixed immovably to the carriage base body **33** by means of their own fastening device **34**. The fixation takes place in particular at the guide housing **28**. Each fixed guide module **25**, **26** is supported on a mounting surface **35** of the carriage base body **33**.
(30) In each guide unit **22**, **23**, the first and second guide modules **25**, **26** are fastened in a working position on the carriage base body **33** by means of the associated fastening device **34** in such a way that a certain pretension is present between the two guide units **22**, **23** and the associated guide surfaces **6**, **7**. Due to the pretension, an at least largely clearance-free adjustment of the guide components can be achieved, resulting in an extremely precise linear movement **17** even with high transverse forces and moments.
(31) Exemplarily, one of the two guide modules **25**, **26** of a respective guide unit **22**, **23** is attached to the carriage base body **33** in an unchangeable working position by means of the associated fastening device **34**. Exemplarily, this is the first guide module **25** in each case, which is therefore also designated as a non-adjustable guide module **25***a*.
(32) Each guide module **25**, **26** has a rear surface **36**, exemplarily formed on the associated guide housing **28**, which is opposed to the associated at least one guide element **27**. Each non-adjustable guide module **25***a* bears with its rear surface **36** against a support section **35***a* of the associated mounting surface **35**, the normal vector of which extends in the transverse direction **18***a* of the carriage.
(33) Forces introduced into the guide elements **27** of the non-adjustable guide modules **25***a* are therefore introduced directly into the carriage base body **33** via the rear surface **36** and the support section **35***a*.
(34) During assembly of the linear guide device **1**, the non-adjustable guide modules **25***a* are placed to the support section **35***a* while the fastening device **34** is still deactivated, and are then immovably fixed by activating the fastening device **34**.
(35) In both guide units **22**, **23**, the pretension is adjusted exclusively by changing the relative position of the second guide module **26** with respect to the carriage base body **33** in the carriage transverse direction **18***a* and thus also with respect to the non-adjustable guide module **25***a*. The second guide module **26** is thus adjustable relative to the carriage base body **33** in the transverse direction **18***a* of the carriage, and therefore is also referred to as adjustable guide module **26***a* for better differentiation.
(36) In the position of use of the guide carriage **15**, the fastening devices **34** associated with the two adjustable guide modules **26***a* are activated, so that the adjustable guide modules **26***a* are also immovably fixed with respect to the carriage base body **33**. The pretension is adjusted when the fastening devices **34** of the adjustable guide modules **26***a* are deactivated, wherein the adjustable guide modules **26***a*, which are no longer fixed, continue to be held on the carriage base body **33**, but are movable with respect to the carriage base body **33** while executing an adjusting movement **37** indicated by a double arrow. The adjusting movement **37** extends in the carriage transverse direction **18***a*.

(37) Preferably, the fastening devices **34** are formed as screw-connecting devices **39**, which applies to the illustrated embodiment example. In this context, each fastening device **34** includes exemplary two clamping screws **34***a* which are arranged at a distance from each other in the carriage longitudinal direction **16***a* and each penetrate, in the carriage vertical direction **19***a*, a fastening bore **40** which is formed in a fastening portion **38** of the carriage base body **33**. A surface of the mounting section **38** facing the underside **14** belongs to the mounting surface **35** and forms a clamping surface **35***b* to which the adjustable guide module **26***a* can be immovably clamped, in particular in a detachable manner, by actuating the clamping screws **34***a*. The clamping screws **34***a* each have a screw head supported on the upper side of the mounting section **38** and a threaded shaft passing through the mounting section **38** and screwed into a threaded hole **34***b* of the guide housing **28** of the adjustable guide module **26***a*.
(38) To adjust the pretension, the screw connections that can be caused via the clamping screws **34***a* are or will be slightly loosened. Thereafter, the adjustable guide modules **26***a* are continuously held on the carriage base body **33**, but are movable in this respect. The mobility results exemplarily among other things from the fact that the fastening bores **40** are formed as elongated holes, the cross-sectional longitudinal axis of which is aligned in the carriage transverse direction **18***a*. The threaded shafts of the loosened clamping screws **34***a* are thus movable in the carriage transverse direction **18***a* together with the adjustable guide module **26***a* held thereon. After the desired adjustment has been made, the screw connections are tightened.
(39) For causing the adjusting movements **37** and accordingly for adjusting the desired pretension, each of the two guide units **22**, **23** is equipped with one of two adjustment units **41**, **42**, wherein the adjustment unit of the first guide unit **22** also being referred to as the first adjustment unit **41** and the adjustment unit of the second guide unit **23** also being referred to as the second adjustment unit **42**.
(40) Each adjustment unit **41**, **42** is capable, when actuated accordingly, of exerting an adjusting force FE oriented in the carriage transverse direction **18***a* on the adjustable guide module **26***a* of the associated guide unit **22**, **23** to cause the adjusting movement **37** of the adjustable guide module **26***a* when the fastening device **34** is deactivated.
(41) By pressing the adjustable guide module **26***a* against the second guide surface **7**, the resulting counterforce causes the associated non-adjustable guide module **25***a* to be simultaneously pressed against the first guide surface **6**. In this way, the two guide modules **25**, **26** of each guide unit **22**, **23** are braced between the two guide surfaces **6**, **7** facing each other.
(42) A characteristic of the linear guide device **1** is that the guide carriage **15** has a central actuating device **43** which is drivingly coupled to both the first adjustment unit **41** and the second adjustment unit **42** and by the actuation of which the adjusting forces FE can be exerted synchronously on the adjustable guide modules **26***a* of the two guide units **22**, **23**.
(43) Since the central actuating device **43** preferably acts directly between the two adjustment units **41**, **42**, in particular without the cooperation of the carriage base body **33**, a force balance can be established with regard to the setting forces FE between the two guide units **22**, **23** to the effect that the setting force FE of the first adjustment unit **41** caused by the central actuating device **43** is at least essentially equal to the setting force FE of the second adjustment unit **42**. This in turn provides the prerequisite for the specification of identical pretensions by the two guide units **22**, **23**.
(44) As can be seen in particular from FIGS. **8** and **9**, the two adjustment units **41**, **42** are preferably each arranged in the transverse direction of the carriage **18***a* between a first guide module **25** and a second guide module **26**.
(45) Preferably, the two adjustment units **41**, **42** are each designed as a wedge gear **44**, which is capable of converting an actuating force FB, generatable via the central actuating device **43** and effective in the longitudinal direction **16** of the carriage, into the desired adjusting force FE, the adjusting force FE acting on the associated adjustable guide module **26***a* orthogonally to the carriage longitudinal direction **16***a*, i.e., in the carriage transverse direction **18***a*.

(46) Preferably, the wedge gear of each adjustment unit **41**, **42** has a movable adjusting element **45** and an immovable adjusting element **46**. The movable adjusting element **45** is movable relative to the carriage base body **33** both in the carriage longitudinal direction **16***a* and in the carriage transverse direction **18***a*. The movement in the carriage longitudinal direction **16***a* is referred to as the driving movement **47** and is indicated by a double arrow. The movement in the carriage transverse direction **18***a*, which is also indicated by a double arrow, is designated as the output movement **48**. The driving movement **47** and the output movement **48** are executed simultaneously in a superimposed manner. In particular, the interaction of the adjusting elements **45**, **46** results in a forced coupling such that each driving movement **47** always simultaneously results in a superimposed output movement **48**.
(47) The driving movements **47** are generatable via the central actuating device **43**, the output movements **48** result from the interaction with the respectively associated immovable adjusting element **46**. The immovable adjusting elements **46** are designed to be stationary, i.e., immovable, relative to the carriage base body **33** in the carriage longitudinal direction **16***a*. Preferably and exemplarily, the immovable adjusting elements **46** are not capable of any relative movement with respect to the carriage base body **33**. In particular, this results from the fact that each immovable adjusting element **46** is a one-piece integral component of the carriage base body **33**. In other words, integral sections of the carriage base body **33** each act as an immovable adjustment element **46**.
(48) In each adjustment unit **41**, **42**, the movable adjusting element **45** is arranged in the carriage transverse direction **18***a* between the adjustable guide module **26***a* and the immovable adjusting element **46**.
(49) Advantageously, there is no fixed connection between the movable adjusting element **45** and the adjacent adjustable guide module **26***a*. The movable adjusting element **45** has a pressing surface **51** facing the adjustable guide module **26***a*, which is in particular a flat surface whose normal vector **52** is orthogonal to the carriage longitudinal direction **16***a* and extends in a plane designated as the adjusting plane **53**, which is spanned by the carriage longitudinal axis **16** and the carriage transverse axis **18**.
(50) The pressing surface **51** constantly abuts a force receiving surface **54** of the adjustable guide module **26***a* facing it, which is formed by its rear surface **36** and whose normal vector **54** also extends orthogonally to the carriage longitudinal direction **16***a* and extends in the adjusting plane **53**.
(51) The driving movement **47** and in this respect the superimposed output movement **48** occur in the adjusting plane **53**.
(52) During its driving movement **47**, the movable adjusting element **45** can slide with the pressing surface **51** against the force receiving surface **54** abutting thereon.
(53) Each movable adjusting element **45** has a first force deflecting surface **56** facing away from the pressing surface **51**, with which it bears slidingly displaceable against a second force-deflecting surface **57** facing it and arranged on the adjacent immovable adjusting element **46**.
(54) It is advantageous that each of the two force deflection surfaces **56**, **57** is formed as an oblique surface inclined to the carriage longitudinal direction **16***a* and to the carriage transverse direction **18***a*. The inclination of the two force deflection surfaces **56**, **57** in contact with each other is identical. The inclination of the oblique force deflection surfaces **56**, **57** with respect to the carriage longitudinal direction **16***a* advantageously has an angle of inclination **60** which lies in an angular range between 5 degrees and 15 degrees and which is preferably 10 degrees. The angle of inclination **60** can be selected such that there is a self-locking between the movable adjusting element **45** and the immovable adjusting element **46**.
(55) The two movable adjusting elements **45** are exemplary wedge-shaped and may therefore also be referred to as wedge elements of the wedge gear **44**.
(56) The force deflection surfaces **56**, **57** are aligned in such a way that their normal vectors extend

in the adjustment plane **53** or in a plane parallel thereto, namely parallel to one another.
(57) The force deflection surfaces **56**, **57** are preferably flat surfaces comparable to so-called inclined planes, although at least one of these two force deflection surfaces **56**, **57** could also have a spherical contour, for example.
(58) In a non-illustrated embodiment, one of the two force deflection surfaces **56**, **57** is formed only punctually or linearly on a projection, in particular a cam-like projection, of the associated adjusting element **45**, **46**. However, the realization of both force deflection surfaces **56**, **57** as mutually parallel inclined surfaces of the same inclination promises the best possible support and force transmission.
(59) When a movable adjusting element **45** executes a driving movement **47**, it slides with its first force-deflecting surface **56** against the second force-deflecting surface **57**, which is stationary relative to the carriage base body **33**, and is displaced in the carriage transverse direction **18***a*, resulting in an output movement **48** superimposed on the driving movement **47**. Since the movable adjusting element **45** consequently displaces in the direction of the adjacent adjustable guide module **26***a*, it exerts with its pressing surface **51** an adjusting force FE on the opposite force receiving surface **54** of the adjustable guide module **26***a*, wherein the adjustable guide module **26***a* and the non-adjustable guide module **25***a* are pressed apart in the carriage transverse direction **18***a* and a desired pretension is established between the guide modules **25**, **25***a*, **26**, **26***a* and the guide surfaces **6**, **7**.
(60) Preferably, the actuating directions of the two wedge gears **44** are opposite to each other in the carriage longitudinal direction **16***a*. This applies to the illustrated embodiment example. This functionality is achieved exemplarily by the fact that the two force deflection surfaces **56**, **57** of the first adjustment unit **41** have an opposite inclination with respect to the carriage longitudinal direction **16***a* as the two force deflection surfaces **56**, **57** of the second adjustment unit **42**. Accordingly, in order to generate the two adjustment forces FE, the two movable adjustment elements **45** are to be driven to oppositely directed driving movements **47**.
(61) Preferably, the inclination of the two pairs of force deflection surfaces **56**, **57** is matched to each other in such a way that in both cases the force deflection surfaces **56**, **57** approach the associated adjustable guide module **26***a* with increasing axial distance from the respective other guide unit **22**, **23**. In other words, the two pairs of force deflection surfaces **56**, **57** move away from the adjustable guide modules **26** with increasing axial approach. As a result, in order to cause the output movement **48**, the two movable adjusting elements **45** must be driven in the sense of moving away from each other axially.
(62) The aforementioned driving of the two movable adjusting elements **45** can be caused via the central actuating device **43**. This is designed in particular and also exemplarily for manual actuation. It includes, i.e., the central actuating device **43**, by way of example, two actuating members **58**, **59**, which are also referred to as first actuating member **58** and second actuating member **59** for better differentiation, and which can be actuated simultaneously or alternatively individually in order to generate an actuating force FB, by means of which the driving movements **47** of the two movable adjusting elements **45** are caused.
(63) The exemplary central actuating device **43** features, for drive coupling with the two movable adjusting elements **45**, a coupling structure designed in the form of a rod and referred to therein as a rod coupling structure **62**, which also includes the two actuating members **58**, **59**. By means of the rod coupling structure **62**, an adjusting force FB with respect to each movable adjusting element **45** can be generated by actuating each of the two actuating members **58**, **59**, the two actuating forces FB pointing away from each other in the carriage longitudinal direction **16***a*. Due to the sole coupling of the movable adjusting elements **45** via the rod coupling structure **62**, at least essentially equal actuating forces FB are constantly generated at the two adjustment units **41**, **42**, so that consequently at least essentially equal adjusting forces FE are also constantly generated.
(64) To realize the rod-coupling structure **62**, each movable adjusting element **45** is exemplarily

penetrated by a threaded hole **63** in the carriage longitudinal direction **16***a*. At least one length portion of the threaded hole **63** features an internal thread **64** in each case, into which one of two screw members **58***a*, **59***a* is screwed, these two screw members **58***a*, **59***a* each forming one of the two actuating members **58**, **59**.

(65) The rod-coupling structure **62** further includes a rigid coupling rod **65** which extends in the carriage longitudinal direction **16***a* between the two movable adjusting members **45**, and its two axial end portions **66** extent into the respective hole of the two threaded holes **63**. A relative mobility between the coupling rod **65** and the two movable adjusting elements **45** is ensured in the carriage longitudinal direction **16***a*.

(66) Preferably, the coupling rod **65** is simultaneously radially and linearly displaceable supported in the threaded hole **63** of each movable adjusting element **45**. The end portions **66** preferably extend in an internally threadless length portion of the respective threaded hole **63** adjoining the internal thread **64**.

(67) Preferably, the coupling rod **65** and the threaded holes **63** have a circular cylindrical contour.

(68) Each coupling rod **65** ends axially with a rod end face **67**, against which the screw member **58***a*, **59***a* screwed into the same threaded hole **63** rests with an inner end face.

(69) Each screw member **58***a*, **59***a* has, on its outer end face axially opposite the coupling rod **65**, a force introduction section **68** at which a torque can be introduced in order to cause a rotary movement **71**, indicated by a double arrow, of the associated screw member **58***a*, **59***a*. The force introduction section **68** is contoured in particular in such a way that a screwing tool can be applied to it. The rotary movement **71** results in a linear actuating movement **72**, illustrated by a double arrow, of the relevant screw member **58***a*, **59***a* relative to the associated movable adjusting element **45** in the carriage longitudinal direction **16***a*.

(70) The pretension for the linear guide can be adjusted by executing the rotary movement **71** of at least one of the two screw members **58***a*, **59***a*. Both screw members **58***a*, **59***a* rest with their inner end face against the facing rod end face **67**. If one of the screw members **58***a*, **59***a* is turned further into the associated threaded hole **63** by manually turning the force introduction section **68** while executing an actuating movement **72**, the coupling rod **65** experiences a pushing force in the direction of the other of the two movable adjusting elements **45**, wherein this other movable adjusting element **45** is pushed away axially outwardly because it is acted upon by the coupling rod **65** via the screw member **58***a*, **59***a* screwed into its threaded hole **63**. At the same time, the rotating movable adjusting element **45** in threaded engagement with the screw member **58***a*, **59***a* is also pressed axially outward because the rotating screw member **58***a*, **59***a* is also supported on the coupling rod **65**.

(71) Overall, there is thus a kind of floating bearing of the two movable adjusting elements **45** within the two adjustment units **41**, **42**, which results in the fact that regardless of which of the two screw members **58***a*, **59***a* undergoes a rotary actuation **71**, both movable adjusting elements **45** are always synchronously exposed to an actuating force FB of the same magnitude, resulting in the desired adjusting forces FE of the same magnitude.

(72) The double arrangement of screw members **58***a*, **59***a* permits convenient adjustment of the pretension from either one of the two axial end faces of the guide carriage **15**.

(73) In a non-illustrated alternative embodiment of a rod coupling structure **62**, only one of the two movable adjusting elements **45** is provided with a screw member **58***a*, **59***a*, the coupling rod **65** being supported on the other of the two movable adjusting elements **45** in a non-adjustable manner in this respect.

(74) Instead of the rod coupling structure **62**, there may also be another coupling structure of the central actuating device **43** that is directly effective between the two movable adjusting elements **45**. For example, a lever structure having pivot levers would be possible.

Claims

1. A linear guide device comprising: a guide carriage featuring a carriage base body which has two guide units arranged in succession in a carriage longitudinal direction, each guide unit having a first guide module arranged on the carriage base body and a second guide module arranged on the carriage base body at a distance opposite to the first guide module in a carriage transverse direction orthogonal to the carriage longitudinal direction, wherein the guide carriage, in its operating position, is supported linearly movably on a base structure in the carriage longitudinal direction, wherein the two first guide modules bear against a first guide surface and the two second guide modules bear against a second guide surface of the base structure so as to be movable with respect thereto, wherein one of the two guide modules of each guide unit is an adjustable guide module, and each of the two guide units is provided with an adjustment unit, the adjustment units being arranged in succession in the carriage longitudinal direction, by means of which the associated adjustable guide module is adjustable by an adjusting force to cause a movement oriented in the carriage transverse direction, relative to the carriage base body, whereby a pretension is adjustable by adjusting the guide modules with the adjusting force with which the two guide modules of the guide unit are pressed against the respectively associated first or second guide surface of the base structure in the operating position of the guide carriage, and wherein the guide carriage features a central actuating device drivingly coupled to the adjustment units of both guide units, through the actuation of which synchronous adjusting forces can be exerted on the two adjustable guide modules.
2. The linear guide device according to claim 1, wherein the two first guide modules and the two second guide modules are designed as rolling-element guide modules with rolling bearing elements provided for bearing against the guide surfaces of the base structure.
3. The linear guide device according to claim 2, wherein each of the two adjustment units is arranged in the carriage transverse direction between the first guide module and the second guide module of the associated first or second guide unit.
4. The linear guide device according to claim 1, wherein each of the two adjustment units is designed as a wedge gear by means of which an actuating force, generatable via the central actuating device and effective in the carriage longitudinal direction, can be converted into an adjusting force acting on the associated adjustable guide module orthogonal to the carriage longitudinal direction.
5. The linear guide device according to claim 4, wherein actuating directions of the two wedge gears are aligned in the carriage longitudinal direction and are oriented opposite to each other.
6. The linear guide device according to claim 1, wherein each of the two adjustment units features a movable adjusting element drivable relative to the carriage base body to perform a driving movement in the carriage longitudinal direction, and an immovable adjusting element fixed with respect to the carriage base body in the carriage longitudinal direction, wherein the movable adjusting element is force-transmittingly coupled to the adjustable guide module for exerting the adjusting force in the carriage transverse direction, and wherein each of the two adjusting elements of each adjustment unit is provided with one of two force deflecting surfaces abutting against each other in a relatively slidable manner, of which at least one force deflecting surface is formed as an oblique surface inclined with respect to the carriage longitudinal direction and to the transverse direction of the carriage.
7. The linear guide device according to claim 6, wherein the movable adjusting element of each adjustment unit features a pressing surface facing the adjustable guide module, which bears against a force-receiving surface of the adjustable guide module supported immovably with respect to the carriage base body in the carriage longitudinal direction and which, during the driving movement of the movable adjusting element, slides against the force-receiving surface while exerting the

adjusting force, oriented in the carriage transverse direction, in the carriage longitudinal direction.
8. The linear guide device according to claim 7, wherein each of the two abutting force deflecting surfaces of each adjustment unit is formed as an oblique surface inclined with respect to the carriage longitudinal direction, wherein the inclination with respect to the carriage longitudinal direction of the two force deflecting surfaces belonging to the same adjustment unit is identical.
9. The linear guide device according to claim 7, wherein the at least one force deflecting surface of one of the two adjustment units has an opposite inclination with respect to the carriage longitudinal direction as the at least one force deflecting surface of the other of the two adjustment units, wherein, in order to generate the adjusting forces, the two movable adjusting elements are drivable by the central actuating device to mutually oppositely oriented driving movements with respect to the carriage longitudinal direction.
10. The linear guide device according to claim 6, wherein each of the two force deflecting surfaces of each adjustment unit is formed as an oblique surface inclined with respect to the carriage longitudinal direction, wherein the inclination with respect to the carriage longitudinal direction of the two force deflection surfaces belonging to the same adjustment unit is identical.
11. The linear guide device according to claim 10, wherein the at least one force deflecting surface of one of the two adjustment units has an opposite inclination with respect to the carriage longitudinal direction as the at least one force deflecting surface of the other of the two adjustment units, wherein, in order to generate the adjusting forces, the two movable adjusting elements are drivable by the central actuating device to mutually oppositely oriented driving movements with respect to the carriage longitudinal direction.
12. The linear guide device according to claim 6, wherein the at least one force deflecting surface of one of the two adjustment units has an opposite inclination with respect to the carriage longitudinal direction as the at least one force deflecting surface of the other of the two adjustment units, wherein, in order to generate the adjusting forces, the two movable adjusting elements are drivable by the central actuating device to mutually oppositely oriented driving movements with respect to the carriage longitudinal direction.
13. The linear guide device according to claim 6, wherein the movable adjusting element of each adjustment unit features a pressing surface facing the adjustable guide module, which bears against a force-receiving surface of the adjustable guide module supported immovably with respect to the carriage base body in the carriage longitudinal direction and which, during the driving movement of the movable adjusting element, slides against the force-receiving surface while exerting the adjusting force, oriented in the carriage transverse direction, in the carriage longitudinal direction, wherein both a normal vector of the pressing surface and a normal vector of the force-receiving surface are oriented in the carriage transverse direction.
14. The linear guide device according to claim 1, wherein the central actuating device is designed to be manually actuated and features at least one actuating member for manual actuation.
15. The linear guide device according to claim 1, wherein the central actuating device features a rod coupling structure for drive coupling with the two adjustment units, with which opposing actuating forces can be introduced synchronously into the two movable adjusting elements of the adjustment units to generate the adjusting forces in the carriage longitudinal direction.
16. The linear guide device according to claim 15, wherein each of the two adjustment units features a movable adjusting element drivable relative to the carriage base body to perform a driving movement in the carriage longitudinal direction, and an immovable adjusting element fixed with respect to the carriage base body in the carriage longitudinal direction, wherein the movable adjusting element is force-transmittingly coupled to the adjustable guide module for exerting the adjusting force in the carriage transverse direction, and wherein each of the two adjusting elements of each adjustment unit is provided with one of two force deflecting surfaces abutting against each other in a relatively slidable manner, of which at least one force deflecting surface is formed as an oblique surface inclined with respect to the carriage longitudinal direction and to the transverse

direction of the carriage and wherein the rod coupling structure features a coupling rod extending between the two movable adjusting elements, the coupling rod is axially supported with respect to each movable adjusting element, wherein a rotatable actuating member of the rod coupling structure cooperating with the coupling rod is arranged on at least one movable adjusting element for axial support, wherein a rotational actuation of the rotatable actuating member changes a distance between the two movable adjusting elements, causing the driving movements of the two movable adjusting elements result.

17. The linear guide device according to claim 16, wherein each movable adjusting element of the rod coupling structure features a threaded hole extending in the carriage longitudinal direction, into which the coupling rod is axially displaceably inserted with one out of two end sections of the could rod, wherein an actuating member formed as a screw member is screwed into each threaded hole, on which the coupling rod with its associated end section is supported on an end face of the end section of the coupling rod and which is axially adjustable by rotation relative to the associated movable adjusting element.

18. The linear guide device according to claim 1, wherein the adjustable guide modules are each stationary fixed to the carriage base body by an activated fastening device, wherein the adjusting movements can be executed with the fastening devices deactivated and the adjustments made can be fixed by activating the fastening devices.

19. The linear guide device according to claim 1, wherein the linear guide has the base structure featuring a longitudinal extension, which has the first guide surface and the second guide surface spaced therefrom, wherein the guide carriage is supported linearly movably on the base structure in the longitudinal direction thereof in that the guide carriage bears movably with the first two guide modules against the first guide surface and with the second two guide modules against the second guide surface.

20. The linear guide device according to claim 1, wherein the two first guide modules and the two second guide modules are designed as rolling-element guide modules with rolling bearing elements provided for bearing against the guide surfaces of the base structure, wherein the two first guide modules and the two second guide modules are designed as recirculating ball guide modules.

21. The linear guide device according to claim 1, wherein the adjustable guide modules are each stationary fixed to the carriage base body by an activated fastening device, wherein the adjusting movements can be executed with the fastening devices deactivated and the adjustments made can be fixed by activating the fastening devices, wherein each of the fastening devices is designed as a screw-connection device which have at least one clamping screw which is tightened when the fastening device is activated and loosened when the fastening device is deactivated.